

Cheuk-Yiu CHAN

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HIGHLIGHT

- **Education:** BEng First Class Honours (CGPA: 4.05/4.3), MPhil in Deep Learning (Feb 2026)
- **Research:** Generative diffusion models, sequence modeling; 3 papers in IEEE TIP/TCE, 1 in ACM MMSports
- **Awards:** Received over 10 prestigious awards including HKSAR Government Scholarship (2019-2021) and HSBC Scholarship for academic excellence
- **Competition:** WorldQuant IQC 2026 participant; WorldQuant Challenge — Gold level (WorldQuant Brain)
- **Current:** Building quant trading system with ML; exploring prediction market alpha detection
- **Writing:** Technical blog <https://allanchan339.github.io/> (vLLM, local LLM, Qwen); reader engagement on Reddit (~218 karma, 53 contributions from write-ups).

EDUCATION

- **BEng in Electronic and Information Engineering** The Hong Kong Polytechnic University
GPA: 4.05/4.3 (First Class Honours) Sep 2017 - Aug 2021
 - **Course:** Computer Vision and Pattern Recognition (A+), surveillance Studies and Technologies (A+), Digital Signal Processing (A+), Mathematics I, II (A+, A+), Image and Audio Processing (A), Probability and Engineering Statistics (A)
- **Master of Philosophy in Electrical and Electronic Engineering** The Hong Kong Polytechnic University
GPA: 3.74/4.3 Sep 2022 - Feb 2026
 - **Research Topic:** Image Enlightening with Deep Learning for Playground Lighting Enhancement with Action Recognition
 - **Supervisors:** Prof. Yui-Lam Chan (MIEEE) and Prof. SIU Wan-Chi (Life-FIEEE)
 - **Funding:** The study is fully supported by Saint Francis University

RESEARCH & TECHNICAL EXPERIENCE

- **ForgeBook: Event-Driven Multi-Venue Quant Stack** Independent
End-to-end quantitative trading & research Feb 2026 – Present
 - **Architecture:** NautilusTrader execution core (open-source, Rust-native event engine; Python strategy layer): event-driven backtest/live parity, streaming replay, and risk checks. ForgeDB (TimescaleDB: compression, indexing, cold/hot tiering, continuous aggregates, vacuum). Exchange-native REST connectors (WebSocket market data planned). vLLM for self-hosted strategy research. ForgeBookAuto (Autoresearch) connects LLM agents to the same engine for automated strategy search. 350+ unit and integration tests (pytest) for ForgeBook; 70+ for ForgeBookAuto.
 - **Universe & data:** 233+ **spot** pairs on the Binance–Bybit–MEXC symbol intersection; 1 m OHLCV from each asset's listing date; automated backfill; TimescaleDB footprint $\approx$ 35 GB compressed.
 - **Statistical analysis:** Feature pipelines for pairs; correlation and cointegration-style screens; volatility modeling for signal design and risk sizing. Cross-sectional pooled backtests supported by NautilusTrader.
 - **ForgeBookAuto (LLM workflow):** Screened 2000+ strategy variants; surfaced 6 candidates passing research gates (in-sample Sharpe ≥ 1 , Calmar ≥ 1 , ≥ 30 trades, exposure time $\geq 5\%$). Closed loop from research prompts $\rightarrow$ generated code $\rightarrow$ deterministic backtests on NautilusTrader; walkthrough: <https://youtu.be/aEvj0SiU6WI>.
 - **Operations:** Spot-only posture with max leverage cap 0.5 $\times$; VPS deployment; Podman containers; exchange REST polling on a 30-minute cadence for OHLCV backfill; 15-minute health checks with Discord alerts.
- **WorldQuant Brain** Virtual
International Quant Championship (IQC) 2026; WorldQuant Challenge (Gold) Apr 2026 – Present
 - **Submissions & rank:** 12 alpha submissions; platform leaderboard #6692 / 131 649 (global, $\approx$ top 5.1% by rank/cohort) and #42 / 1697 among Hong Kong participants ($\approx$ top 2.5%); snapshot—ranks move with cohort.
 - **Signal families:** Price momentum and mean reversion; options-implied volatility; sentiment; fundamentals. Engineering emphasizes cross-sectional comparability, volatility-aware scaling, and turnover control matched to each family's horizon.
 - **IQC 2026 Metric:** WorldQuant Brain *combined* in-sample aggregate for submitted Stage 1 alphas: Sharpe 3.51, fitness 2.30, turnover 29.1%, returns 12.45%, max drawdown 1.66%.
- **Research Assistant** Saint Francis University
GPU cluster, procurement, research infrastructure Sep 2021 – Sep 2025
 - **Research Infrastructure:** Purchased and managed GPU server cluster (12+ nodes); maintained electricity and air conditioning systems for stable computing environment
 - **Procurement:** Handled hardware purchase tender processes from evaluation to approval; evaluated specifications including NVIDIA A100 GPUs with NVLink InfiniBand switches for high-performance cluster
- **Generative Diffusion Models with Learnable Latent Variables** The Hong Kong Polytechnic University
MPhil research, Department of Electrical and Electronic Engineering Feb 2023 – Aug 2025

- **Research Focus:** Proposed novel generative diffusion models for data synthesis with mathematical innovations. Key contributions include: (1) Anchoring Diffusion Probabilistic Models - introduced learnable anchor mean $\mathcal{N}(m_t, I)$ extending standard $\mathcal{N}(0, I)$ noise, providing additional degrees of freedom for better data distribution modeling; developed Dynamical Regulated Diffusion Anchoring (DRDA) mechanism to anchor the enhancement process and ensure fidelity to input; (2) Back Projection Generative Strategy - developed degradation-aware diffusion framework with emphasis on statistical fidelity modeling between normal/low-light domains, enabling high-quality synthetic data generation for robust model training. Created LOL-Diff, a large-scale synthetic dataset to address data scarcity in low-light image enhancement tasks.

- **Publication:**

- [1] **C.-Y. Chan**, W.-C. Siu, Y.-H. Chan, and H. A. Chan, "AnlightenDiff: Anchoring Diffusion Probabilistic Models on Low Light Image Enhancement", in IEEE Transactions on Image Processing, vol. 33, pp. 6324-6339, 2024, doi: 10.1109/TIP.2024.3486610.
- [2] **C.-Y. Chan**, W.-C. Siu, Y.-H. Chan, and H. A. Chan, "Back Projection Generative Strategy for Low and Normal Light Image Pairs With Enhanced Statistical Fidelity and Diversity," in IEEE Transactions on Consumer Electronics, vol. 71, no. 2, pp. 3575-3586, May 2025, doi: 10.1109/TCE.2024.3516366.
- [3] **C.-Y. Chan**, W.-C. Siu, Y.-H. Chan, and H. A. Chan, "Generative Strategy for Low and Normal Light Image Pairs with Enhanced Statistical Fidelity," 2024 IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 2024, pp. 1-3, doi: 10.1109/ICCE59016.2024.10444437.

• Spatiotemporal Trajectory Modeling for Action Prediction

The Hong Kong Polytechnic University

Graduate research, Department of Electrical and Electronic Engineering

Jan 2022 – Dec 2022

- **Research Focus:** Developed sequence-to-sequence models for multi-object trajectory prediction in sports video analytics. Built state estimation systems to track player and ball positions across continuous video streams and forecast future events (goals, offsides, fouls) in real-time. Proposed novel attention mechanisms for capturing long-range temporal dependencies and multi-scale feature fusion. Achieved sub-second latency for real-time event prediction, demonstrating feasibility for live broadcast applications.
- **Publication:**
 - [4] **C.-Y. Chan**, C.-C. Hui, W.-C. Siu, S.-w. Chan, and H. A. Chan, "To Start Automatic Commentary of Soccer Game with Mixed Spatial and Temporal Attention," in TENCON 2022-2022 IEEE Region 10 Conference (TENCON), Nov. 2022, IEEE.
 - [5] W.-C. Siu, H. A. Chan, S.-W. Chan, W.-H. Cheng, B.-C. Yang, **C.-Y. Chan**, C.-C. Hui, and R. Salahudeen, "On the completion of automatic football game commentary system with deep learning," in Proc. SPIE International Workshop on Advanced Imaging Technology (IWAIT), Mar. 2023, pp. 125920H, doi: 10.1117/12.2668398.
 - [6] S. Giancola et al. (**C.-Y. Chan**), "SoccerNet 2022 Challenges Results," in Proceedings of the 5th International ACM Workshop on Multimedia Content Analysis in Sports, in MMSports '22., 2022, pp. 75–86. doi: 10.1145/3552437.3558545

HONORS, AWARDS & SCHOLARSHIPS

- Gold Level, WorldQuant Challenge (WorldQuant Brain) — 2026
- HSBC Greater Bay Area (Hong Kong) Scholarship (The Hongkong and Shanghai Banking Corporation) — 2021
- ASM Pacific Technology Scholarship (The Hong Kong Polytechnic University) — 2021
- Mitsubishi Electric (Hong Kong) Limited Scholarship (The Hong Kong Polytechnic University) — 2021
- Best Academic Performance Award (The Hong Kong Polytechnic University) — 2021
- Outstanding Student Award (The Hong Kong Polytechnic University) — 2021
- Hong Kong Special Administrative Region Government Scholarship (Hong Kong SAR Government) — 2019–2021
- HSBC Hong Kong Scholarship (The Hong Kong Polytechnic University) — 2020
- Mitsubishi Electric (Hong Kong) Limited Scholarship (The Hong Kong Polytechnic University) — 2020
- VTech Group of Companies Scholarship (The Hong Kong Polytechnic University) — 2020
- Gold Award, Be a Global Citizen Campaign (Chinese YMCA of Hong Kong) — 2019
- Dr. Ng Tat-lun Memorial Scholarship (The Hong Kong Polytechnic University) — 2018
- Dean's Honours List (The Hong Kong Polytechnic University) — 2018–2021
- Best (Semester) GPA Award (The Hong Kong Polytechnic University) — 2017–2021

RESEARCH INTERESTS

- **Machine Learning:** Deep Learning, Generative Models, Sequence Modeling, Harness Engineering (agent orchestration, verification loops)
- **Applications:** Financial Signal Extraction, Time-series Prediction, Statistical Modeling, Prediction Markets

TECHNICAL SKILLS

- **ML/AI:** PyTorch, Diffusion Models, Transformers, LLMs
- **Mathematics:** Probability Theory, Bayesian Statistics, Linear Algebra, Optimization
- **Trading & Programming:** Python, SQL, TimescaleDB, NautilusTrader, scikit-learn, LaTeX
- **Infrastructure:** GPU cluster (12+ nodes), Linux, Git, Podman, Tailscale, Cloudflare
- **Languages:** English (fluent); Chinese (bilingual); Cantonese (native)